Szymon Sarnowicz

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Experience

Readiscover.app – Technical Lead | *Jan 2026 to present*

- Built full-stack semantic search platform for long-form text retrieval with adaptive chunking and multi-layer caching (GCS for persistent storage, Redis for sub-second cache access with optimized TTL).
- Implemented LLM query reformulation (Gemini 2.5 Flash at temperature 0.0 for deterministic output) before embedding, prioritizing trustworthiness by returning only continuous verbatim passages and eliminating hallucination risk (vs. 43.6% for GPT-5.1).
- Conducted rigorous ablation testing across chunking parameters and evaluated against GPT-5.1 on 50 test questions achieving 34% top-3 accuracy with zero hallucinations and superior proximity to correct passages, with results published at University of Michigan and deployed as production platform.

General Motors – Software Engineer II | *Feb 2023 to present*

- Owned end-to-end redesign of safety-critical fault detection system in C/C++ with sensor data processing, signal filtering algorithms, and state machine logic, architecting persistent state management to prevent critical failures and help avoid ~$480M in potential recalls.
- Architected and deployed production-level diagnostic system across distributed services with state monitoring logic and alert routing, increasing system reliability through refactored fault-response behavior and a Python testing framework for safety-critical validation.

Data4Good – University of Michigan – Lead Developer, Hangul Project | *Dec 2023 to Dec 2024*

- Scoped and designed Svelte web application for United Nations document summarization with UN representatives, defining requirements through iterative reviews to ensure product alignment with real-world workflows, ultimately shipping to production and automating 100% of document summarization tasks while shifting workflow to review-only.
- Developed application using Apache Tika for HTML parsing and BART model for NLP-based extraction, redesigning the API pipeline into stages with exponential backoff to increase success rate from 53% to 92%.

General Motors – Program Manager, Autonomous Systems (Ultra Cruise) | *Jul 2021 to Feb 2023*

- Tripled data collected per sprint by automating data-request workflows, improving visibility for perception and autonomy teams and enabling faster iteration cycles for sensor fusion and real-time decision-making pipelines in autonomous vehicle development.
- Authored comprehensive troubleshooting guides and diagnostics workflows that enabled cross-functional teams to independently resolve camera, radar, and LiDAR system issues while bridging hardware integration and software systems, improving sensor calibration diagnostics and reducing development cycle time.

General Motors – Software Engineer (Rotational Program) | *Sep 2019 to Jul 2021*

- Automated calibration data management processes across embedded systems, eliminating 1 hour of daily manual data entry while gaining foundational experience in embedded systems, real-time data handling, sensor calibration, and automotive software development practices.

Education

Master's in Applied Data Science, University of Michigan, summa cum laude | *Dec 2022 to Dec 2025*
B.S. Mechanical Engineering, University of Washington, cum laude | *Sep 2015 to Jun 2019*

Technical Skills & Awards

Programming Languages: Python, C, C++, JavaScript, SQL
Systems: C++ for embedded systems, real-time signal processing, state machines
ML/AI: LLM integration, semantic search, vector embeddings, NLP, RAG
Cloud & Infrastructure: GCP (Vertex AI, Cloud Run), AWS (Lambda, DynamoDB), Docker, Redis
Tools: React, FastAPI, Git, Linux
Best Cybersecurity Hack, MHack16 (350+ participants) | Dec 2023
Native Languages: English, Polish